

HARI L

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EDUCATION

Saveetha Engineering College (Autonomous), Chennai / *B.E. Mechanical Engineering* 2025 – 2029

- **Coursework:** Thermodynamics, Fluid Mechanics, Mechanics of Solids, Manufacturing Technology, CAD/CAM, Heat & Mass Transfer, Machine Design, Metrology

PUBLICATIONS

- **Edge-Based Predictive Diagnostics for Industrial Monitoring Systems** — SAE International Automotive & Mobility APAC Conference 2026 (*accepted*). Edge sensor fusion and anomaly detection with emphasis on latency, offline reliability and fault-escalation logic.

PROJECTS

OILIQ — Real-Time Equipment Condition Monitoring

- Built a multi-sensor system tracking temperature, soot, oxidation and TBN with real-time Normal / Caution / Critical classification and fault-escalation logic, at sub-5 ms processing latency.
- Handled sensor integration, wiring, calibration and hardware validation under real operating constraints; reached Round 2 of Caterpillar Tech Challenge 2026.

Predictive Maintenance Edge System — ASME IMECE Brain Bolt 2026

- Designed a three-sensor edge system (ADXL355 vibration, PT100 temperature, current transformer) on ESP32-S3 for centrifugal pump predictive maintenance.
- Qualified for the South Regional round and National Finale; team worked on welding distortion in pressure vessel fabrication with ANSYS thermal simulation.

Smart India Hackathon 2026 — Mechanical Design & Additive Manufacturing 2026

- Developed a robotic assembly in Fusion 360 using parametric modelling, Generative Design and structural FEA; optimized for FDM printing.
- Fabricated a 0.25x scale prototype for design validation; produced renders, simulation visuals

AMR Fleet Simulation — Autonomous Mobile Robotics

- Built a multi-robot fleet simulation covering normal operation, blockage, robot failure, communication loss and stress scenarios.
- Extended it from a single process to a multi-process architecture with independent robot workers and UDP communication, including packet serialization and packet-loss handling, validated through repeatable tests.

Industrial Software Tools

- **CAD Validation Tool:** Automated pre-check for thin walls and geometric inconsistencies, with machine-readable reports for manufacturing handoff.
- **Fleet Monitoring Dashboard:** Multi-unit health scoring, threshold alerts and fault escalation.
- **Audit Trail Logger:** labeled event pipeline for root-cause analysis and corrective-action records.

TECHNICAL EXPERIENCE

Open-Source Contributor — **dora-rs Robotics Framework** 2025 – Present

- Contributed reviewed production code (~861 lines): SigNoz/OTLP observability instrumentation and TFLite hardware-inference integration.

LEADERSHIP & ACHIEVEMENTS

- **Caterpillar Tech Challenge 2026:** Team OILIQ reached Semi Finale (top 30 of 1,500+ teams).
- **ASME IMECE Brain Bolt:** 2026 National Finalist (Top 15 of 300+ Teams) ; 2025 Finalist (CFD Automation Sprint).
- **Treasurer - ASME Student Chapter**
- **SAE INDIA** - Design and Simulation Lead (BAJA SAEINDIA) , Vice-Captain (SAEINDIA Bicycle Design Challenge 8th Edition)

TECHNICAL SKILLS

CAD & Design: Fusion 360, Generative Design, DfAM, Design for Manufacturing, 3D Printing (FDM)

Simulation: ANSYS (Fluent/PyFluent), Structural FEA, CFD, MATLAB, CAD Validation

Programming & Tools: Python, C/C++, Git, GitHub, Linux, WSL2